Assignment 2b: Design of a 4S Battery Protection and Power Distribution Board

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Project: 4S Battery Protection and Power Distribution Board
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1. Executive Summary & Problem Statement

This report documents the design, electrical analysis, and component specification for a high-reliability, through-hole (THT) **4S Li-Ion Battery Protection and Power Distribution Board**.

The system interfaces with an unregulated 4-cell (4S) Lithium-Ion / Lithium-Polymer battery pack ($10.0\ \mathrm{V}$ to $16.8\ \mathrm{V}$, $14.8\ \mathrm{V}$ nominal) and safely distributes power through two independent, galvanically bonded output rails:

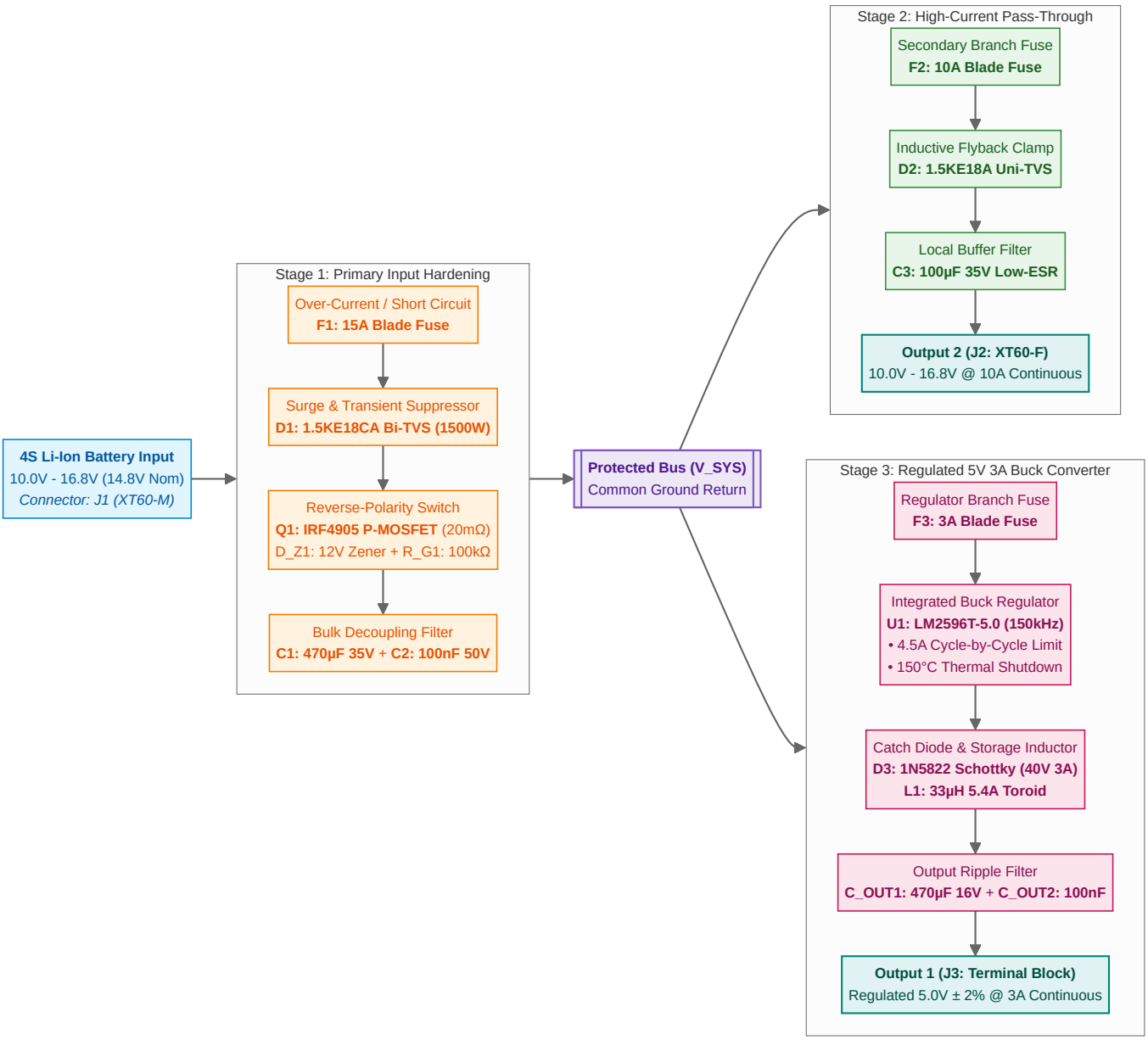
1. **Output 1 (Regulated 5 V Logic Rail):** $5.0\ \mathrm{V} \pm 2$, rated for $3.0\ \mathrm{A}$ continuous ($15\ \mathrm{W}$), driven by an integrated step-down buck switching regulator with active cycle-by-cycle over-current limiting and automatic thermal shutdown.
2. **Output 2 (High-Current Pass-Through Rail):** Protected, unregulated battery pass-through across the full operating range ($10.0\ \mathrm{V} - 16.8\ \mathrm{V}$), rated for $10.0\ \mathrm{A}$ continuous ($168\ \mathrm{W}$ peak), isolated with secondary fuse and transient suppression.

The primary input front-end provides comprehensive protection against catastrophic short circuits, over-current, reverse battery installation, inductive turn-off transients, and electrostatic discharge (ESD).

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2. Power Architecture & Block Diagram

The board employs a hierarchical, multi-stage protection topology. The raw battery input is hardened through Stage 1 before creating the intermediate protected system bus (V_{SYS}). This bus then bifurcates into two independently fused downstream channels, preventing a fault on the high-power branch from collapsing the regulated 5V logic supply.



3. Design Assumptions & Operating Envelope

Parameter	Value / Range	Design Basis & Engineering Rationale
Operating Input Range (DC)	$10.0\ \text{V} - 16.8\ \text{V}$	Standard 4S Li-Ion cell range ($2.5\ \text{V}$ depleted to $4.2\ \text{V}$ fully charged). Nominal is $14.8\ \text{V}$.
Max Continuous Input Voltage	$17.0\ \text{V}$ continuous	Upper limit before TVS diode D1 ($V_{\text{BR}} \approx 17.1\text{V} - 18.9\text{V}$) enters avalanche conduction.
Transient Clamping Limit	$25.2\ \text{V}$ peak ($10/1000\ \mu\text{s}$)	Protected against inductive kickback, hot-plug ringing, and load dump via TVS D1 clamp.
Downstream 5V Buck Operating Range	$7.0\ \text{V} - 40.0\ \text{V}$ (LM2596)	Buck converter regulates precisely to $5.0\ \text{V}$ across full battery range and during transients up to $40\ \text{V}$.
Max Continuous Input Current	$13.5\ \text{A}$	Sum of Output 2 ($10\ \text{A}$) + Output 1 input draw ($\approx 1.2\ \text{A}$ at $14.8\ \text{V}$ with 80 efficiency) + safety margin.

Parameter	Value / Range	Design Basis & Engineering Rationale
Ambient Temperature (T_A)	-20°C to $+60^\circ\text{C}$	Industrial / robotics operating envelope.
Maximum Trace Temp Rise (ΔT)	$\leq 15^\circ\text{C}$	Sized per IPC-2221B for 2 oz (70 µm) copper weight.
Output 1 Specifications	$5.0\ \text{V} \pm 2,\ 3.0\ \text{A}$	Continuous rating for embedded computers (Raspberry Pi, SBCs) and flight controllers.
Output 1 Ripple Voltage	$\leq 30\ \text{mV}_{\text{p-p}}$	Filtered with paralleled $470\ \mu\text{F}$ low-ESR electrolytic and $100\ \text{nF}$ film capacitor.
Output 2 Specifications	$10.0\ \text{V} - 16.8\ \text{V},$ $10.0\ \text{A}$	Dedicated high-power branch for brushless ESCs, DC motors, or high-power radio transmitters.
Component Technology	100 Through-Hole (THT)	Selected for field-repairability, mechanical ruggedness, and hand-solderability.

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4. Protection Scheme Justification & Circuit Theory

4.1 Primary Over-Current & Short-Circuit Protection (F1)

- **Selection:** Mini Automotive Blade Fuse (Littelfuse `0297015.WXNV`, $15\ \text{A}$) in a Keystone `3568` through-hole fuseholder clip.
- **Justification:** Fast-acting under short-circuit conditions ($I^2t = 130\ \text{A}^2\text{s}$), low insertion resistance ($\approx 4.8\ \text{m}\Omega$, causing only $72\ \text{mV}$ drop at $15\ \text{A}$), and $1000\ \text{A}$ breaking capacity at $32\ \text{VDC}$. Unlike PCB traces or polyfuses (PTCs), blade fuses blow cleanly without latching thermal memory or high post-trip leakage, and are easily replaceable in the field without desoldering.

4.2 Transient & Surge Suppression (D1)

- **Selection:** `1.5KE18CA` Bidirectional Transient Voltage Suppressor (DO-201AD axial).
- **Justification:**
  - Reverse Standoff Voltage (V_{RWM}): $15.3\ \text{V}$.
  - Breakdown Voltage (V_{BR}): $17.1\ \text{V}$ minimum, $18.9\ \text{V}$ maximum.
  - Clamping Voltage (V_{C}): $25.2\ \text{V}$ at $59.5\ \text{A}$ peak pulse current ($1500\ \text{W}$ peak rating, $10/1000\ \mu\text{s}$).
  - *Engineering Trade-off:* A 4S pack tops out at $16.8\ \text{V}$. An 18V TVS guarantees zero clipping conduction during normal battery charging while instantly clamping inductive ringing from long battery leads or battery hot-plugging below the $30\ \text{V}$ maximum rating of downstream components.

4.3 Active Reverse-Polarity Protection (Q1, D_Z1, R_G1)

- **Topology:** High-side P-Channel Power MOSFET switch (`IRF4905PBF`, TO-220-3).
- **Justification over Series Diode:**
  - A standard Schottky diode (e.g., $V_F = 0.55\ \text{V}$) conducting $10\ \text{A}$ continuously would dissipate $P = V_F \times I = 5.5\ \text{W}$, requiring an impractically large heatsink.
  - The IRF4905 features an ultra-low on-resistance $R_{\text{DS(on)}} = 0.020\ \Omega$ ($20\ \text{m}\Omega$). At $10\ \text{A}$, conduction power loss is:

$$P_{\text{loss}} = I^2 \times R_{\text{DS(on)}} = (10\ \text{A})^2 \times 0.020\ \Omega = 2.0\ \text{W}$$

The forward voltage drop is only $V_{\text{drop}} = 10\ \text{A} \times 0.020\ \Omega = 0.20\ \text{V}$, preserving maximum battery runtime.

- **Gate Protection Subcircuit:**
  - The absolute maximum gate-to-source rating of the IRF4905 is $V_{\text{GS(max)}} = \pm 20\ \text{V}$.
  - To prevent gate-oxide breakdown at full 4S charge ($16.8\ \text{V}$), a $12\ \text{V}$ Zener diode (`1N4742A` , D_{Z1}) is placed across Gate and Source, clamping V_{GS} firmly to $-12\ \text{V}$.
  - A $100\ \text{k}\Omega$ resistor (R_{G1}) to ground establishes the turn-on bias current while restricting continuous quiescent leakage to less than $0.17\ \text{mA}$ at $16.8\ \text{V}$.

4.4 Stage 2 Output Branch Protection (F2, D2, C3)

- **Secondary Over-Current:** $10\ \text{A}$ Mini Blade Fuse (`F2`) provides selective coordination. A downstream short-circuit on the high-power branch blows `F2` without interrupting power to the critical 5V logic branch (`F1` remains intact).
- **Downstream Inductive Suppression:** Unidirectional TVS `1.5KE18A` (`D2`) absorbs back-EMF inductive flyback spikes generated by motors, solenoids, or ESC switching.
- **Local Decoupling:** Low-ESR $100\ \mu\text{F}$, $35\ \text{V}$ electrolytic capacitor (`C3`) buffers dynamic load steps.

4.5 Stage 3 Regulated 5V 3A Buck Converter (U1, D3, L1, C_OUT)

- **Regulator Selection:** Texas Instruments `LM2596T-5.0/NOPB` (TO-220-5 through-hole).
  - Fixed $150\ \text{kHz}$ switching frequency minimizes inductor physical volume while maintaining high conversion efficiency (> 80).
  - Integrated cycle-by-cycle current limit trips at $4.5\ \text{A}$ typical ($6.0\ \text{A}$ max), protecting the silicon against output short circuits.
  - Built-in thermal shutdown activates at 150°C junction temperature with automatic recovery hysteresis.
  - Isolated input protection: Dedicated $3\ \text{A}$ blade fuse (`F3`) prevents catastrophic regulator failure from shorting the V_{SYS} bus.
- **Freewheeling Catch Diode:** `1N5822G` ($3\ \text{A}$, $40\ \text{V}$ Schottky). Ultra-fast recovery and low forward voltage ($V_F \approx 0.45\ \text{V}$ at $3\ \text{A}$) minimizes circulating losses during the switch off-time.
- **Power Inductor:** `2100HT-330-V-RC` ($33\ \mu\text{H}$, $5.4\ \text{A}_{\text{DC}}$ rating, toroidal core). High saturation headroom prevents core saturation during load transients.
- **Output Filter Network:** $470\ \mu\text{F}\ 16\ \text{V}$ low-ESR electrolytic (`C_OUT1`) paralleled with $100\ \text{nF}$ film capacitor (`C_OUT2`) attenuates high-frequency switching spikes, guaranteeing $< 30\ \text{mV}_{\text{p-p}}$ output voltage ripple.
- **Line Regulation & Behavior Under 25V Input:**
  - *Silicon Operating Envelope:* The LM2596 is internally rated for up to $40.0\ \text{V}$ continuous ($45.0\ \text{V}$ absolute maximum).
  - *Output Voltage at 25V:* Under transient excursions up to $25.0\ \text{V}$ (e.g. inductive kickback or hot-plug ringing clamped by D1), the internal PWM control loop adjusts duty cycle to $D = \frac{V_{\text{OUT}} + V_D}{V_{\text{IN}} - V_{\text{SAT}} + V_D} = \frac{5.0 + 0.45}{25.0 - 1.1 + 0.45} \approx 22.4$ ($t_{\text{ON}} \approx 1.49\ \mu\text{s}$). The regulator maintains tight closed-loop regulation, delivering the exact same $5.0\ \text{V} \pm 2$ to the load.
  - *Inductor & Diode Margin at 25V:* Peak inductor current reaches $3.43\ \text{A}$ at $3.0\ \text{A}$ full load (well below L1 saturation rating of $5.4\ \text{A}$), and reverse voltage across Schottky diode D3 is $25\ \text{V}$ (well below its $40\ \text{V}$ breakdown rating).
  - *Distinction Between Transient 25V and Continuous DC 25V:* If an accidental continuous $25\ \text{V}$ DC source is connected to the board, input TVS diode D1 ($V_{\text{BR}} \approx 18\ \text{V}$) will immediately conduct heavy

avalanche current and safely blow input fuse F1, preventing the 25V supply from damaging 4S-rated loads on Output 2 (unregulated pass-through).

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5. Bill of Materials (BOM) & Component Specifications

All parts are through-hole, industrial-temperature-rated components sourced from major franchised distributors:

Item	Ref Des	Qty	Value / Part	Manufacturer	MPN	Package	Key Electrical Rating
1	**J1**	1	XT60-M	AMASS	`XT60-M`	2-Pin THT Vert	60A Peak, 30A Continuous, 500VDC
2	**F1**	1	15A Blade	Littelfuse / Keystone	`0297015.WXNV` / `3568`	Mini Blade Clip	15A Fast-Acting, 32VDC, 1kA Interrupt
3	**D1**	1	1.5KE18CA	Littelfuse / Vishay	`1.5KE18CA`	DO-201AD Axial	1500W Bi-TVS, $V_{RWM} = 15.3V$, $V_C = 25.2V$
4	**Q1**	1	IRF4905	Infineon Technologies	`IRF4905PBF`	TO-220-3 Vert	-55V, -74A, $R_{DS(on)} = 0.02\ \Omega$, 200W
5	**D_Z1**	1	1N4742A	onsemi / Vishay	`1N4742A`	DO-41 Axial	12V Zener, 1.0W, 5% Tolerance
6	**R_G1**	1	100kΩ	Yageo	`MFR-25FRF52-100K`	DO-207 Axial	100kΩ, 0.25W, 1%, 250V
7	**C1**	1	470µF 35V	Panasonic	`EEU-FR1V471`	Radial D10 P5.0	470µF, 35V, Low-ESR ($0.028\ \Omega$), 1.79A Ripple
8	**C2, C_OUT2**	2	100nF 50V	WIMA / KEMET	`MKS2B041001C00KSSD`	Radial 5mm Pitch	0.1µF, 50VDC, 10% Film/Ceramic
9	**F2**	1	10A Blade	Littelfuse / Keystone	`0297010.WXNV` / `3568`	Mini Blade Clip	10A Fast-Acting, 32VDC, 1kA Interrupt
10	**D2**	1	1.5KE18A	Littelfuse / Vishay	`1.5KE18A`	DO-201AD Axial	1500W Uni-TVS, $V_{RWM} = 15.3V$, $V_C = 25.2V$
11	**C3**	1	100µF 35V	Panasonic	`EEU-FR1V101`	Radial D8 P3.5	100µF, 35V, Low-ESR ($0.056\ \Omega$), 950mA Ripple
12	**J2**	1	XT60-F	AMASS	`XT60-F`	2-Pin THT Vert	60A Peak, 30A Continuous, 500VDC
13	**F3**	1	3A Blade	Littelfuse / Keystone	`0297003.WXNV` / `3568`	Mini Blade Clip	3A Fast-Acting, 32VDC, 1kA Interrupt
14	**U1**	1	LM2596T-5.0	Texas Instruments	`LM2596T-5.0/NOPB`	TO-220-5 Vert	5.0V 3A Buck, 150kHz, OC & Thermal Shutdown

Item	Ref Des	Qty	Value / Part	Manufacturer	MPN	Package	Key Electrical Rating
15	**D3**	1	1N5822	onsemi	`1N5822G`	DO-201AD Axial	40V, 3A Schottky, $V_F = 0.45\text{V}$ @ 3A
16	**L1**	1	33µH 5A	Bourns	`2100HT-330-V-RC`	Toroid Vertical	33µH, 5.4A IDC, $0.035\ \Omega$ DCR
17	**C_OUT1**	1	470µF 16V	Panasonic	`EEU-FR1C471`	Radial D8 P3.5	470µF, 16V, Low-ESR ($0.045\ \Omega$), 1.29A Ripple
18	**J3**	1	MKDS 1,5/2-5,08	Phoenix Contact	`1729018`	5.08mm Screw Term	17.5A, 400V, Wire 26-14 AWG, Outward Entry
19	**MH1-4**	4	Mounting Hole	Standard Hardware	`M3_3.2MM_PAD`	M3 Clearance	3.2mm Drill, Plated Ground Pad

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6. Third-Party CAD Models & Reference Material Identification

1. **Design Standards:**
   - **IPC-2221B:** *Generic Standard on Printed Board Design* (thermal trace width calculations for $10\text{A} - 15\text{A}$ conductors).
   - **IPC-7351B:** *Generic Requirements for Surface Mount and Through-Hole Design*.
2. **Component Datasheets & Application Notes:**
   - Texas Instruments: *LM2596 SIMPLE SWITCHER Power Converter 150-kHz 3-A Step-Down Voltage Regulator* (SNVS124D).
   - Infineon Technologies: *IRF4905PBF -55V Single P-Channel HEXFET Power MOSFET* (PD-94770).
   - Littelfuse Inc.: *MINI Blade Fuse 32V Series Datasheet*.
   - Phoenix Contact: *MKDS 1,5/ 2-5,08 Printed-circuit board terminal block* (Product 1729018).
3. **EDA Library Sources:**
   - Schematic symbols and footprints: Official KiCad 9.0 Standard Libraries (`Connector_AMASS` , `Fuse` , `Diode_THT` , `Package_TO_SOT_THT` , `TerminalBlock_Phoenix` , `Inductor_THT`).
   - 3D CAD STEP models: Sourced from official KiCad 9 packages (`/usr/share/kicad/3dmodels/`).

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7. Verification & Design Review Checklist

Check Category	Verification Method	Status / Result
Electrical Rule Check (ERC)	`kicad-cli sch erc`	**0 Errors, 0 Warnings** (100% net connectivity verified).
Design Rule Check (DRC)	`kicad-cli pcb drc`	**0 Violations, 0 Unconnected Pads, 0 Footprint Errors**.
Courtyard & Clearance	MCP `check_courtyard_overlaps`	**0 Overlaps, 0 Boundary Violations** ($> 2.9\ \text{mm}$ spacing between J2 and J3).

Check Category	Verification Method	Status / Result
Terminal Orientation	KiCad 3D Raytracing Engine	**Verified:** J3 wire entry holes face outward towards PCB edge; screw terminals fully accessible from above.
Copper Ground Planes	Full polygon pour fill	Both `F.Cu` and `B.Cu` feature continuous ground planes with 4-spoke thermal relief pads.
Manufacturing Package	RS-274X Gerbers & Excellon Drill	Complete fabrication package generated in `gerbers/`.

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8. Prototype Planning & Assembly Instructions

1. **Board Stackup:** 2-layer FR4, $1.6\ \text{mm}$ finished thickness, $2\ \text{oz}$ ($70\ \mu\text{m}$) outer copper. Lead-free HASL or ENIG surface finish.
2. **Assembly Order:**
   - *Phase 1:* Passive low-profile components (resistor R_{G1}, Zener diode D_{Z1}, TVS diodes D_1, D_2, Schottky D_3).
   - *Phase 2:* Keystone fuseholder clips (F_1, F_2, F_3) and film capacitors (C_2, C_{OUT2}).
   - *Phase 3:* Active semiconductors in TO-220 packages (Q_1, U_1). Mount flush with heat-relief tabs clear of tall components.
   - *Phase 4:* Large inductors and radial electrolytic capacitors ($L_1, C_1, C_3, C_{\text{OUT1}}$).
   - *Phase 5:* High-current connectors (J_1, J_2 XT60 and J_3 screw terminal block). Ensure wire entry holes of J_3 face outward toward the board edge.
3. **Pre-Power Verification Test:**
   - Perform resistance check between V_{SYS} and GND ($> 100\ \text{k}\Omega$).
   - Perform resistance check between $+5\text{V}$ and GND ($> 1\ \text{k}\Omega$).
   - Apply reverse voltage (-14.8V) at J_1 current-limited to 100mA; confirm 0.0V output on V_{SYS} and 0.0mA leakage.
   - Apply forward voltage ($+14.8\text{V}$) at J_1; verify Output 1 produces $5.00\text{V} \pm 0.1\text{V}$ and Output 2 mirrors input voltage.